

Registration

Speaker 1

[00:13]

Good morning everyone, and thank you for taking the time to watch my presentation.

My name is Yasmin Ferrera, and today I will be presenting my statistical analysis of the Pima Indians Diabetes dataset using R.

[00:56]

Diabetes is one of the most common chronic diseases worldwide and represents a major public health challenge. Understanding the factors associated with diabetes can help identify individuals at higher risk and support earlier intervention. According to the American Diabetes Association (2024), risk factors such as age, obesity and elevated blood glucose play an important role in the development of Type 2 diabetes.

In this presentation, I will first introduce the dataset and explore its main characteristics using descriptive statistics and visualisations. I will then perform inferential statistical analyses, including hypothesis testing, correlation analysis and regression modelling. Finally, I will discuss the main findings, limitations of the study and recommendations for future research.

[00:45]

The analysis is based on the Pima Indians Diabetes Dataset, a well-known benchmark dataset in healthcare analytics. It contains information from 768 female participants and includes eight clinical and demographic predictors alongside a binary outcome indicating the presence or absence of diabetes. These variables provide the foundation for both exploratory and predictive analyses. You'll also notice a small R logo on some slides, which indicates that the corresponding R code used for the analysis is included in the hidden section. This keeps the slides clean while still demonstrating how each result was produced.

[00:26]

The analysis followed a structured workflow beginning with data exploration and descriptive statistics. Visualisations were created using ggplot2 to identify patterns and potential anomalies before applying inferential statistical tests and regression models. This approach follows Anscombe's recommendation that numerical summaries and graphical analysis should be used together to understand data effectively.

[00:30]

This slide presents the main characteristics of the study sample. The dataset contains 768 participants, of whom 500 (65.1%) were classified as non-diabetic and 268 (34.9%) as diabetic. This indicates a moderate class imbalance, which is common in medical datasets and is important to consider when interpreting the results of predictive models.

[00:32]

Table 1 summarises the descriptive statistics for the principal variables included in the analysis. The average participant was approximately 33 years old, with a mean BMI of 32 kg/m². Blood glucose levels exhibited notable variability, reflecting differences in metabolic health among participants. The table also highlights the presence of zero values in some clinical variables, a known characteristic of this dataset that should be considered during interpretation.

[00:32]

The age distribution indicates that most participants were relatively young adults. The median age was 29 years, and half of the participants were between 24 and 41 years of age. The histogram demonstrates a positively skewed distribution, meaning there are fewer older participants. The boxplot also identifies several observations above approximately 65 years as potential outliers, although these values remain clinically plausible.

[00:32]

These histograms illustrate the distribution of BMI and glucose levels in the dataset. BMI values are concentrated around 30–35 kg/m², indicating that many participants fall within the overweight or obese range. Glucose values exhibit greater variability and a slight positive skew, reflecting differences in

metabolic status among participants. Both variables also contain zero values, which are likely to represent missing clinical measurements.

[00:30]

The boxplots illustrate the variability of blood pressure and the number of pregnancies within the study population. Blood pressure values are centred around 72 mmHg, although several observations with a value of zero indicate likely missing clinical measurements. The pregnancy variable is positively skewed, with most participants reporting relatively few pregnancies and a small number exhibiting substantially higher values.

[01:21]

This slide presents the diabetes prevalence across different age groups and BMI categories. Looking first at the age groups, the percentage of participants diagnosed with diabetes increased considerably with age. The prevalence rose from approximately 21% in participants under 30 years old to over 46% in the 30–39 group, reaching its highest value of 55% in the 40–49 age group. Although the prevalence was slightly lower in the 50+ group, it remained substantially higher than in younger participants. A similar pattern was observed across BMI categories. Participants with a normal BMI had the lowest diabetes prevalence, at around 6%, whereas almost 46% of obese participants were diagnosed with diabetes. This suggests that increasing BMI is strongly associated with a higher prevalence of diabetes. These descriptive findings provide an initial indication that both age and BMI are important factors related to diabetes. In the next section, these observations are formally tested using inferential statistical methods.

[00:29]

This boxplot compares glucose levels between participants with and without diabetes. The diabetic group shows noticeably higher glucose values, with a mean of approximately 141 mg/dL compared with 110 mg/dL in the non-diabetic group. A Welch two-sample t-test confirmed that this difference is statistically significant ($p < 0.001$), suggesting that glucose is strongly associated with diabetes in this dataset.

[01:33]

On this slide, I examined whether the number of pregnancies differs between women with and without diabetes.

The boxplot shows that women diagnosed with diabetes generally had a higher number of pregnancies than those without diabetes. This is also reflected in the summary statistics. Women without diabetes had an average of 3.30 pregnancies, while women with diabetes had a higher average of 4.87 pregnancies. The median also increased from 2 pregnancies in the non-diabetic group to 4 pregnancies in the diabetic group. To determine whether this difference was statistically significant, I performed an independent Welch's t-test. The test produced a p-value smaller than 0.001, indicating that the difference between the two groups is statistically significant.

In addition, I categorised pregnancies into four groups: 0, 1–2, 3–4, and 5 or more pregnancies. The bar chart shows that the proportion of diabetes increases as the number of pregnancies increases. Nearly half of the women with five or more pregnancies were diagnosed with diabetes, compared with much lower rates in the other groups.

Overall, these findings suggest that a higher number of pregnancies is associated with an increased prevalence of diabetes within this dataset.

[00:44]

Pearson's correlation coefficients were calculated to examine the relationships between the predictor variables and diabetes outcome. Glucose demonstrated the strongest positive association with diabetes ($r = 0.47$), followed by BMI ($r = 0.29$) and Age ($r = 0.24$). Blood Pressure and Skin Thickness showed very weak correlations. Importantly, no correlation exceeded 0.70, indicating that multicollinearity is unlikely to be a concern. These findings support the inclusion of multiple predictors in the subsequent regression analyses.

[00:50]

To investigate group differences, two Chi-square tests and a one-way ANOVA were conducted. The Chi-square analyses demonstrated significant associations between diabetes status and both age group and BMI category, indicating that diabetes prevalence varies across these groups. The ANOVA further showed that mean glucose levels differed significantly by age group. Tukey's post-hoc analysis revealed that participants

aged 50 years and older had significantly higher glucose levels than all younger groups, while there was no significant difference between the 30–39 and 40–49 age groups.

[01:36]

To identify the factors associated with glucose levels, I developed a multiple linear regression model using Age, BMI, Pregnancies, Blood Pressure, Skin Thickness, Insulin, and Diabetes Pedigree Function as predictor variables.

The overall model was statistically significant, with an F-statistic of 32.46 and a p-value below 0.001. The model explained approximately 23% of the variation in glucose levels, as indicated by an R-squared value of 0.23.

Looking at the individual predictors, Insulin was the strongest predictor of glucose levels and showed a highly significant positive relationship. Age and BMI were also significant positive predictors, indicating that older individuals and those with a higher BMI tended to have higher glucose levels.

Interestingly, Skin Thickness showed a significant negative association with glucose levels, while Diabetes Pedigree Function had a smaller but still statistically significant positive effect.

On the other hand, Pregnancies and Blood Pressure were not statistically significant after accounting for the other variables in the model. This suggests that their independent contribution to predicting glucose levels was limited.

Overall, these findings indicate that glucose levels are influenced by multiple clinical factors, with insulin, age, and BMI providing the greatest predictive value in this dataset.

[01:31]

Next, I developed a logistic regression model to predict diabetes status using BMI, age, and glucose as predictors.

All three variables were statistically significant, indicating that they each contribute to predicting the likelihood of diabetes.

The odds ratios show that increasing BMI, age, and glucose are all associated with higher odds of being diagnosed with diabetes. For example, each one-unit increase in BMI increases the odds of diabetes by approximately 8.5%, while each one-unit increase in glucose increases the odds by about 3.3%.

To evaluate the model, I used the Hosmer–Lemeshow goodness-of-fit test. Since the p-value was 0.349, which is greater than 0.05, there is no evidence of poor fit, suggesting that the model fits the data well.

The model achieved an overall classification accuracy of 76.8%. It performed particularly well at identifying non-diabetic individuals, with a specificity of 87.8%, while its sensitivity for identifying diabetic individuals was 56.3%.

Overall, these results indicate that BMI, age, and especially glucose are useful predictors of diabetes risk.

[01:04]

Finally, I investigated whether age modified the effect of BMI on diabetes risk by comparing two logistic regression models. Model 1 included the main effects of BMI, age, glucose, and pregnancies, while Model 2 additionally included the interaction term between BMI and age.

A likelihood ratio test was used to compare the two models. The test was not statistically significant, with a p-value of 0.843, indicating that adding the interaction term did not improve model performance.

The interaction term itself was also not significant, suggesting that the effect of BMI on diabetes risk does not depend on age in this dataset. Therefore, the simpler main-effects model was preferred because it provides a similar fit while being easier to interpret.

Overall, glucose and pregnancies remained significant predictors of diabetes, whereas the BMI-by-age interaction did not contribute additional predictive value.

[01:05]

The findings of this study are consistent with previous research on diabetes risk factors. Age and BMI were both significantly associated with diabetes, while glucose emerged as the strongest predictor in the regression models. These results agree with established evidence that increasing age, obesity, and elevated glucose levels are major contributors to Type 2 diabetes.

Interestingly, the interaction between BMI and age was not statistically significant, suggesting that these variables contribute independently rather than modifying each other's effects in this dataset.

This project also highlights the importance of combining statistical tests with appropriate visualisations. As noted by Anscombe, both calculations and graphical exploration contribute to a better understanding of data.

Overall, the analyses demonstrate how descriptive statistics, inferential testing, and predictive modelling can be integrated to investigate complex healthcare datasets.

[00:36]

This study has several limitations. The dataset contains zero values for variables such as glucose, BMI, blood pressure, and insulin, which are unlikely to represent true clinical measurements. In addition, the data are cross-sectional, meaning we can identify associations but not causation. The models also excluded lifestyle factors that may influence diabetes risk. Future work should focus on improved data preprocessing, additional predictors, more advanced machine learning methods, and validation using external datasets.

[01:06]

In conclusion, this project analysed data from 768 individuals to investigate factors associated with diabetes. Approximately 35% of participants had diabetes. The descriptive and inferential analyses showed that glucose levels and the number of pregnancies differed significantly between diabetic and non-diabetic individuals. Regression analyses further demonstrated that age, BMI and glucose were the strongest predictors of diabetes risk, while the logistic regression model achieved an overall classification accuracy of 76.8%. Overall, this project demonstrates how statistical analysis and predictive modelling can be applied to healthcare data to identify important risk factors and support evidence-based decision-making.

Thank you very much for your time and attention. I hope this presentation has provided a clear overview of the analysis and its findings. I appreciate the opportunity to present my work, and I hope you have found it both informative and insightful. Have a great day.